PRANVEER SINGH INSTITUTE OF TECHNOLOGY

InternsElite

MINOR PROJECT REPORT

ON

STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

Branch: Computer Science and Engineering (Artificial Intelligence & Machine Learning)

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SESSION: 2026–2027 | SEPTEMBER BATCH

1. INTRODUCTION AND PROBLEM STATEMENT

1.1 Introduction

Student performance plays an important role in understanding the academic progress of students. Learning abilities, study habits, attendance, and personal circumstances can affect academic results.

The Student Performance Analysis and Prediction System is a machine learning-based project developed to analyze students' academic data and predict their expected examination scores.

The project uses Python and libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The dataset contains study time, attendance percentage, previous grades, sleep hours, internet access, parental education, extracurricular activities, and other attributes.

Linear Regression is used to predict examination scores. The model is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).

1.2 Problem Statement

Traditional evaluation often focuses on examination marks and may not fully explain the factors associated with academic results. Manually analyzing study hours, attendance, previous performance, and lifestyle attributes for many students can be difficult. This project uses data analysis and machine learning to identify patterns and estimate examination scores.

2. PROJECT OBJECTIVES

1. To collect and understand student academic performance data.
2. To clean and preprocess the dataset by handling missing values and checking duplicates.
3. To perform Exploratory Data Analysis (EDA).
4. To study the relationship between academic and lifestyle attributes and examination scores.
5. To develop a Linear Regression model.
6. To train and test the model using prepared data.
7. To evaluate the model using MAE, RMSE, and R^2.
8. To compare actual and predicted scores.
9. To generate useful insights about student performance.

3. DATASET DESCRIPTION AND SOURCE

3.1 Dataset Description

The dataset contains 1,000 student records and 12 attributes. It includes numerical and categorical information related to academic performance and study habits.

Attribute	Description
student_id	Unique student identifier
study_time_hours	Daily study time in hours
attendance_percent	Attendance percentage
previous_grade	Previous academic grade
final_exam_score	Final examination score; target variable
final_grade	Final grade obtained
gender	Gender category
parental_education	Parents' education level
internet_access	Internet availability
extracurricular_activities	Participation in extracurricular activities
sleep_hours	Average sleeping hours
part_time_job	Whether the student has a part-time job

3.2 Dataset Source

The project uses the CSV file student_performance_dataset.csv stored in the data folder. The dataset was supplied with the project work. No external publisher or URL was specified in the available project material, so no external source is claimed.

4. DATA PREPROCESSING AND FEATURE ENGINEERING

4.1 Data Preprocessing

10. The CSV dataset was loaded using Pandas.

11. The dataset was inspected using head(), info(), and describe().
12. The parental_education column contained 102 missing values (10.2%); these were filled with the mode, High School.
13. Duplicate records were checked; no duplicate records were found.
14. The cleaned data was verified and no missing values remained.

4.2 Feature Engineering

15. student_id was removed because it is an identifier.
16. final_exam_score was selected as the target variable.
17. final_grade was excluded from input features to avoid target leakage.
18. Categorical variables were converted into numerical columns using pandas.get_dummies() with drop_first=True.
19. The data was split into 80% training and 20% testing sets using train_test_split.

5. EXPLORATORY DATA ANALYSIS AND VISUALIZATIONS

Exploratory Data Analysis (EDA) was performed to understand data distributions and relationships between student attributes and examination performance.

5.1 Dataset Overview

- Total records: 1,000 students.
- Total attributes: 12 columns.
- Average study time: approximately 3.57 hours.
- Average attendance: approximately 85.09%.
- Average final examination score: approximately 83.54.
- Maximum final examination score: 100.

5.2 Analysis Performed

A correlation heatmap was used to inspect relationships among numerical variables. Regression plots were created for study time versus final score and attendance versus final score. A scatter plot examined sleep hours against final score.

Group-wise comparisons were performed for final grades, part-time jobs, internet access, extracurricular activities, and parental education. Average final scores by grade were approximately: A – 96.07, B – 84.75, C – 75.41, D – 66.23, and F – 56.68.

6. MACHINE LEARNING METHODOLOGY

This is a supervised learning task because the dataset contains a known numerical target, final_exam_score.

6.1 Linear Regression

Linear Regression is a supervised learning algorithm used to estimate a continuous numerical value. It models the relationship between input features and a target variable using a linear equation. In this project, the input features include study time, attendance, previous grade, sleep hours, and encoded categorical attributes.

6.2 Workflow

20. Load and inspect the dataset.
21. Clean missing values and verify data quality.
22. Perform EDA and prepare visualizations.
23. Separate input features (X) and target (y).
24. Encode categorical input features.
25. Split data into training and testing subsets.
26. Train the Linear Regression model.
27. Predict test-set scores and evaluate the results.

7. MODEL TRAINING AND TESTING APPROACH

7.1 Training

The dataset was divided into 80% training data and 20% testing data using random_state=42. The Linear Regression model was fitted using the training features and corresponding examination scores.

7.2 Testing

After training, the model predicted scores for the unseen testing features. Predicted values were compared with actual scores to evaluate performance.

7.3 Prediction Function

A reusable prediction function was created in src/model.py. It loads the saved model and feature-column list, converts a student's input into a DataFrame, applies one-hot encoding, aligns the columns with the training features, and returns a predicted score rounded to two decimal places. A sample prediction produced 94.08. This is an example prediction, not a guaranteed examination result.

8. EVALUATION METRICS AND RESULTS

The model was evaluated using MAE, RMSE, and R-squared (R^2).

- MAE measures the average absolute difference between actual and predicted scores. Lower values indicate smaller average errors.
- RMSE is the square root of the average squared prediction error and penalizes larger errors more strongly.
- R^2 indicates the proportion of target variation explained by the model.

Metric	Obtained Value
MAE	4.984581
RMSE	6.415737
R^2 Score	0.617588

The MAE is approximately 4.98 score points and RMSE is approximately 6.42 score points. The R^2 score of approximately 0.618 indicates that the model explains about 61.76% of the variation in test-set examination scores.

9. ACTUAL VERSUS PREDICTED EXAMINATION MARKS

An Actual vs Predicted Scores scatter plot was generated for the test dataset. A diagonal reference line represents perfect predictions. Points closer to this line indicate predictions closer to actual scores.

The graph was saved as actual_vs_predicted.png in outputs/visualizations. Prediction results were also saved as a CSV file. For one sample student, the actual score was 71.8 and the predicted score was 78.05. Individual predictions can differ; evaluation metrics summarize performance across the test set.

10. INTERPRETATION OF FINDINGS AND ACADEMIC PERFORMANCE PATTERNS

The EDA results show that academic performance can be examined across study-related, academic, and personal attributes. The average final score was approximately 83.54, and grade-wise averages decreased from A to F.

The model achieved an MAE of approximately 4.98 and an R^2 score of approximately 0.618 on the test data. This indicates that Linear Regression captures some patterns in the dataset, while some score variation remains unexplained. These are associations in this dataset and do not establish that any single factor causes a change in examination scores.

11. LIMITATIONS AND FUTURE IMPROVEMENTS

11.1 Limitations

- The dataset contains 1,000 records and may not represent every student population.
- Linear Regression may not capture complex non-linear relationships.
- Some factors affecting academic performance may not be included.
- Predictions are estimates and should not be treated as actual examination results.
- Detailed external dataset provenance was not available in the supplied project material.

11.2 Future Improvements

- Use a larger and more diverse dataset.
- Compare Linear Regression with Decision Tree and Random Forest regression models.
- Apply cross-validation and hyperparameter tuning.
- Add relevant academic features where reliable data is available.
- Develop a user-friendly web interface for score prediction.
- Add dashboards and explanations for model outputs.

12. CONCLUSION

The Student Performance Analysis and Prediction System was developed to analyze student-related data and estimate final examination scores using machine learning. The work included data inspection, preprocessing, EDA, feature encoding, model training, testing, and evaluation.

The Linear Regression model achieved MAE 4.984581, RMSE 6.415737, and R^2 0.617588 on the test data. A reusable prediction function was implemented and tested with a sample student record. The project demonstrates how data analysis and machine learning can be used to study academic performance patterns. It can be improved with more data, additional algorithms, and an interactive application.

13. REFERENCES

- Pandas Documentation: https://pandas.pydata.org/docs/
- NumPy Documentation: https://numpy.org/doc/
- Matplotlib Documentation: https://matplotlib.org/stable/
- Seaborn Documentation: https://seaborn.pydata.org/
- Scikit-learn Documentation: https://scikit-learn.org/stable/
- Python Documentation: https://docs.python.org/3/